

Tingle Just for You: A Preliminary Study of AI-based Customized ASMR Content Generation

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ABSTRACT

With the recent surge in interest in autonomous sensory meridian response (ASMR), numerous contents that stimulate the user's visual and auditory senses by using various triggers are consumed a lot. However, due to individual differences, people have different reactions to triggers, therefore, have to manually search for the content that suits their preferences. In this paper, we present an AI-based ASMR content generation service and conduct a preliminary user study to figure out its feasibility and usability. Through a user study with 42 participants, we find that the AI-based customized ASMR content generation service could be used to satisfy users. However, we could also learn about various issues to be addressed for better user experience and satisfaction, such as audio-video synchronization, appropriateness, level of noise, and so on.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; *Empirical studies in collaborative and social computing*.

KEYWORDS

ASMR, Sound generation, Generative model, AI-created Contents

ACM Reference Format:

JiYeon Oh, Daun Kim, Jae-Yeop Jeong, Elizaveta Lukianova, and Jin-Woo Jeong. 2023. Tingle Just for You: A Preliminary Study of AI-based Customized ASMR Content Generation. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3544549.3585872>

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CHI EA '23, April 23–28, 2023, Hamburg, Germany
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ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3585872>

1 INTRODUCTION

Autonomous sensory meridian response (ASMR) is a sensory phenomenon that is generally induced in response to audio-visual stimuli, typically depicted as a tingling sensation on the scalp and neck. The tingles, aroused by the response to auditory and visual triggers such as whispering, role-playing, crisp sounds, or slow and repetitive movements, are known to be accompanied by a feeling of relaxation and sleep improvements[1]. There have been many attempts to demonstrate the effectiveness of this phenomenon[3, 7, 10, 18, 19], but there is still a lack of research that provides scientific explanations and evidence. Nevertheless, ASMR has been becoming popular content with the advent of online communities of people who love the ASMR experience and the spreading influence of social media platforms. Especially on YouTube, ASMR creators (called ASMRtists) provide numerous videos to expand viewers' ASMR experience through various sound and visual stimulations. The videos tagged with 'ASMR' on YouTube in 2017 was 4.3 million, a significant increase from 2 million in 2016, and an average of 11 new videos are posted with the tag "ASMR" every hour[16].

In general, ASMRtists use a variety of triggers to satisfy those who want a tingling sensation and engage them. However, the triggers affect individuals differently, so some may experience ASMR very well while others may see no effect or dislike them[1, 9, 10, 12]. Therefore, users need to search for ASMR content appealing to their tastes by themselves to achieve the best experience for ASMR. With the recent development of generative AI technologies, there has been a surge of interest in multimedia content generated by AI. With generative deep learning models, users can easily obtain the desired high-quality result by telling AIs what they want to take. For example, by just typing specific prompts or keywords, people can generate attribute-controlled sentences [4], get solution codes for programming exercises, change the language of the original programming code to another one [8, 20], and even create an image or video [5, 13–15, 17].

Inspired by the recent advances in generative AI approaches, in this work we propose an AI-based solution that generates ASMR content that best matches the user's preferences and flavor. This

paper is a preliminary study to investigate the feasibility of AI-generated ASMR content, so that we particularly focus on exploring the quality of the generated ASMR content and the user responses to it.

2 USER STUDY

Despite the increase in supply and demand for ASMR content, there was not much HCI research conducted in this field. Existing studies focused on the analysis of user reactions to real ASMR content or aimed at characterizing ASMR triggers based on the user's bio-signal such as a change in the pupil, EEG, and heart rate [3, 12, 19] and YouTube comments [11]. The popularity of ASMR continues to rise in media culture, and the impact of AI-created media will be spreading as well. Therefore, unlike the previous studies that have investigated ASMR content created by humans, we examine the user's response to AI-created ASMR content and evaluate its usability.

2.1 Research Questions

In this study, we defined the following questions to figure out how users think about ASMR content generated by AI.

- **RQ1: Does AI-generated ASMR content satisfy users?** In order for AI-generated ASMR sounds to satisfy users, first of all, they must sound like audio in actual ASMR content. Furthermore, not only auditory but also visual stimuli mainly trigger ASMR and since human brain can inherently recognize the correspondence of audio-visual modalities, they must sound suitable for the visual information of the video. Hence, to address our first research question, we explore if the level of user satisfaction with AI-generated ASMR audio vs real ASMR audio would be different. Afterward, we examine how visual cues affect users' assessment by comparing the user responses obtained when visual+audio stimuli are provided vs only audio stimuli are provided. In addition, we also investigate whether providing sound unrelated to visual information degrades user satisfaction.
- **RQ2: What do users expect from AI that generates customized ASMR?** The final goal of our study is to develop an AI service that creates ASMR content that suits the user's preference and purpose. Therefore, to find out the feasibility of the user-customized AI-based ASMR generation system and the future direction of our research, we investigate the user's responses and expectations towards a pilot version of our AI system that takes user input text to generate ASMR video.

2.2 AI-based ASMR Content Generation

In this section, we discuss how we create and organize ASMR content for our experiments. As a base dataset for AI-infused ASMR generation, we used the YouTube-ASMR dataset presented by [21], originally designed as a study for matching the sound source location in the visual stream with spatial information in the audio stream. This dataset contains over 900 hours of ASMR video clips from 300K unique videos collected from YouTube, each of which is 10 seconds long on average. Since sounds imitating human speech is much more difficult to generate, we did not use ASMR clips that

are mainly based on ASMRtists' voices, such as role-playing content where whisper-like conversations are the main triggers.

With reference to the ASMR video categories analyzed in [2, 11], we selected 13 ASMR videos consisting of various multimodal interactions, such as approaching the viewer (i.e., cam) with their hands or tools, brushing someone's hair, touching the microphone, or tapping and scratching a certain object. Fig. 1 shows the example screenshot of the selected videos used in our experiment.

To generate an ASMR sound suitable for the selected videos, we used SpecVQGAN [6], one of the state-of-the-art cross-modal autoregressive samplers, that receives a sequence of visual features and generates visually related and high-fidelity sound. Fig. 2a represents how we generate ASMR content using a generative deep learning model. As shown in the figure, an original ASMR video is delivered to the SpecVQGAN model as input first. Then, SpecVQGAN analyzes the visual information of the given video data and generates an audio stream that could fit the visual content. Finally, the visual information of the original video and the AI-created sound is merged to complete a new ASMR content. In this stage, we only create an audio part for the given original ASMR video content. It should also be noted that the SpecVQGAN model only can produce mono audio, unlike the YouTube ASMR videos including stereo audio. Since the spatial feature of audio may affect the evaluation of user satisfaction with the real content and the generated content, the audio of the YouTube ASMR content was changed to mono. Finally, the generated ASMR content and mono-converted real ASMR content are trimmed equally to 3 seconds.

2.3 Experimental Procedure

The experiment was performed with an online survey which includes a set of queries regarding our research questions. First, we randomly chose 7 ASMR content pairs to ask the following question: "which content do you think is more satisfying or more natural?", which is to verify whether and how participants perceive AI-generated ASMR audio and real ASMR audio differently. Fig. 1 represents a set of ASMR content used for this question. For this question, the participants were given two sets of ASMR content, where the first set contains audio-only content while the second set contains audio+visual content. We asked the participants to answer the same question for each content set to figure out if the visual stimuli affect the user's decision. Additionally, to see how the provision of sound unrelated to the visual information affects user satisfaction with ASMR content, we also compared the participants' evaluations towards i) a video containing plausible sound generated by AI and ii) a video containing totally unrelated sound generated by AI. Fig. 2b depicts a procedure to generate ASMR content with visually unrelated sound. Another six ASMR content not used in the previous question were used to prepare two sets of ASMR videos for this question.

The survey was conducted online to obtain answers from varied age groups, and a total of 42 people (26 females and 16 males, 30 in 10-20s, and 12 in 30-50s) participated. Among them, 29 people never listen to ASMR content at all, and the rest 13 people listen to ASMR content at least once a week. The survey consisted of mix-up questions, a 7-point Likert scale for user satisfaction-related questions, and the optional short-answer questions, and took approximately

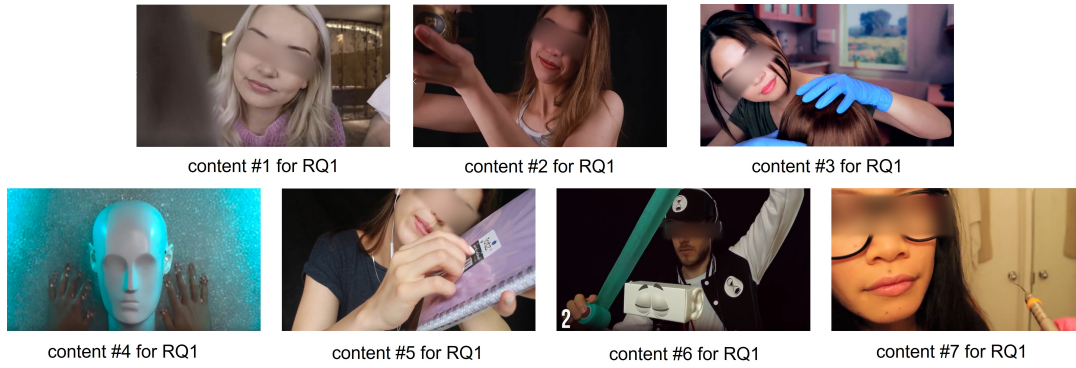


Figure 1: Examples of videos used in this study

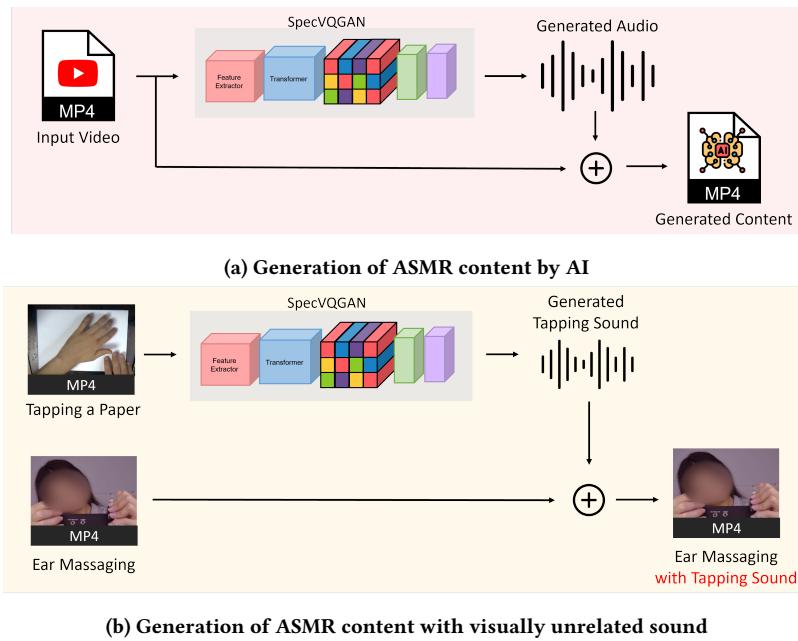


Figure 2: Framework to Generate ASMR content

10 minutes on average. The participants were instructed to answer the survey while watching the ASMR content we provide related to each question. To remove any bias or effect on the judgment of satisfaction towards ASMR content (i.e., RQ1), we never included any statement on the presence of AI-generated ASMR media.

3 RESULT AND ANALYSIS

3.1 RQ1: Does AI-generated ASMR contents satisfy users?

Comparison of AI-generated and Real ASMR Content Fig.3 shows the result for the question "which content do you think is more satisfying or more natural?". The y -axis represents the percentage of people who selected AI-generated ASMR content. Therefore, the value 50 indicates that no difference in preference for a certain ASMR content was found. Similarly, a value >50 means

that AI-created content was preferred. The participants tended to select real content more than AI-generated ones, and when visual information was provided, the number of participants who preferred real ASMR content was increased. However, these trends did not apply to the four contents below:

(1) Case 1: participants preferred the AI-generated content when only audio was provided.

Content #1 Since the main trigger of this content was wiping the camera with a tissue, the sound volume was too low. Therefore, the AI-generated sound may have been chosen because of its more intensive acoustic stimulation.

Content #7 In the video, an ASMRtist is digging ears with a metal earpick. However, the actual recorded audio did sound like a sharp metal banging. Therefore, when only the audio was given, more people preferred the AI-generated content

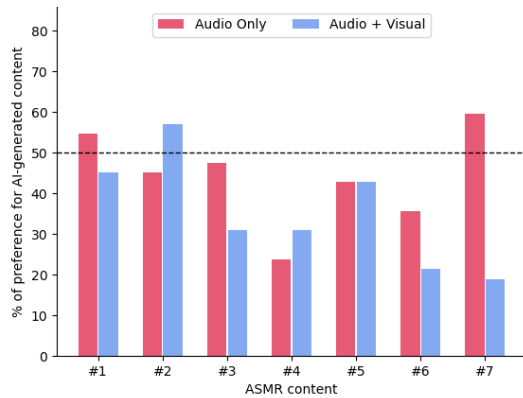


Figure 3: Ratio of preference for AI-generated content

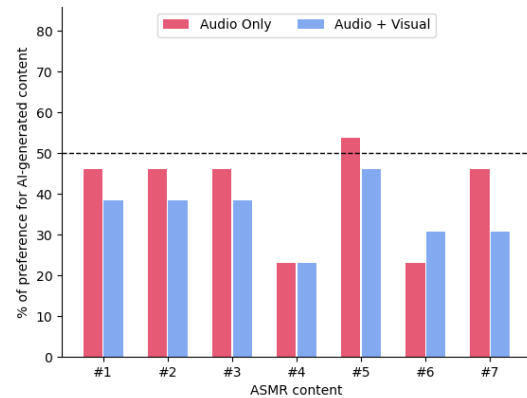
because it sounds rather softer. However, when the visual cue is provided together, the participants have chosen the real content more because they found out that the sound was caused by the metal.

(2) Case 2: participants' satisfaction on the AI-generated content increased even more when visual information was provided together.

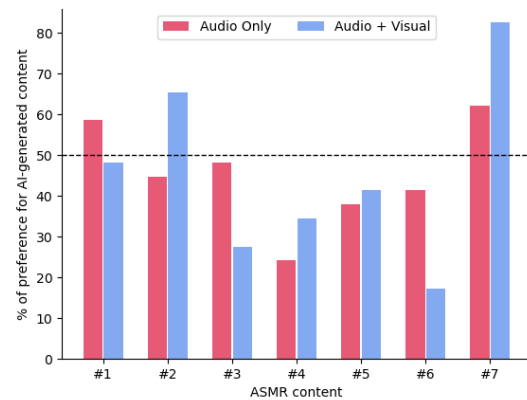
Content #2 A singing bowl is a tool commonly used for meditation-type ASMR content, ASMRtists use vibrations caused by tapping or rubbing the surface as the main stimulus. Since 69% of the participants said they never listen to ASMR content at all, they might not have any idea of where this kind of sound comes from. This confusion could make the participants feel like the AI-generated sound is more appropriate even though visual information is provided.

Content #4 This video contains audiovisual stimulation of touching an orbeez. The participants preferred the real sound which provides more auditory stimulus when hearing audio-only content. However, when the visual information was presented, some of them moved to the AI-generated content which made a muffled sound as if being in the water.

As discussed above, there could be a difference in the evaluation between those who do have knowledge of ASMR triggers and their corresponding sound, and those who do not. Therefore, we divided the users into two groups according to the level of ASMR usage and experience: i) ASMRViewer (people who listen to it at least once a week) and ii) nonASMRViewer (people who do not listen to ASMR at all), and then examined if there exists any difference in the preference. Fig. 4 depicts the difference in preference between the group. As shown in the figure, in most cases the ASMRViewer group preferred real content. On the other hand, the nonASMRViewer group seemed they were overall unable to distinguish between the AI-generated and real audio content. In particular, there was a big difference between the group in the case of content #2 and #7 described above. Specifically, the nonASMRViewer group preferred the AI-generated content much more. From these observations, we could validate that the assessment of ASMR content would differ



(a) Group of People Who Listen More than Once a Week



(b) Group of People Who Do Not Listen at All

Figure 4: Ratio of preference for AI-generated content: Comparison based on ASMR experience

depending on whether one has any previous experience with ASMR content.

Impact of Visual Information In Fig. 3, it is worth noting that in the case of an audio-only set, the difference between the ratio of participants who preferred real content versus those who preferred AI-generated content was not that significant. Given that participants were not aware of the presence of AI-created content when answering this question, it could be interpreted that participants cannot easily distinguish between real and AI-generated content when visual information is not provided. This also could be confirmed by the participants' comments. When the participants were asked: "Between Audio-only vs Audio+visual content set, which side did you find it easier in making a decision?", 69% of the participants said it was easier to choose when visual materials were provided together. We believe that if visual information is provided, people could have an explicit evaluation cue like whether it is well synchronized and/or well matched with visual information, resulting in easier decision-making. Some participants who chose content

with visual information mentioned that *"Visual data was used as a sort of comparative criterion!"* and *"When only hearing the audio, it was difficult to make a decision since no additional information was provided"*. On the contrary, one participant who chose content with audio-only responded like *"The visual information made it more confusing :("*.

Therefore, we conducted an extra survey to figure out which factors mainly affected the participant's evaluation in case visual information is provided together. As a result of the survey, 71.4% of the participants said their choice was affected by the synchronization issue between audio and video, and 76.19% mentioned that mismatch between the actual audio content and their expectation was another important point. Additionally, several participants responded that the 'decrease in the resolution of sound due to noise' and 'the magnitude of sound' also affected the overall evaluation. The survey for audio-only and audio+visual content can be found from <https://sites.google.com/view/asmr-audio-only/> and <https://sites.google.com/view/asmr-with-visual/>, respectively.

Effect of Sound Unrelated to Visual Information on User Satisfaction In order to evaluate how AI-generated sound unrelated to visual information affects user satisfaction, we also prepared a series of AI-generated ASMR content which contains an audio stream that is unrelated to the visual information of the video. This consisted of three videos, each of which is 3s long. The participants were instructed to watch both ASMR video sequences i) with appropriate sound content and ii) with unrelated sound content, and then answer to question "How much are you satisfied with these contents?" as a Likert-7 scale point (1: never satisfied, 7: very satisfied). The generated videos with appropriate sound and with unrelated sound can be found from <https://youtu.be/gPscBFHd6t0> and <https://youtu.be/g6PcfH0re3I>, respectively. The results of the survey for this question are shown in Fig. 5. The participants' satisfaction on the videos with i) appropriate sound content and ii) unrelated sound content were 2.95 and 4.3 on average, respectively. The result indicates that the audio-visual correspondence significantly affects user satisfaction with the ASMR content. Some participants said, *"One of the videos was difficult to listen to because extraneous sounds also could be heard besides the expected sounds, while the other video was satisfactory because the expected sound was heard without any noise."* and *"One video sounds more like a noise than ASMR content"*. From this, we could confirm that the more the sound is inconsistent with the visual stimuli, the more people feel that the audio is noisy or not pleasant.

3.2 RQ2: What do users expect from AI that generates customized ASMR?

For RQ2, we conducted an additional survey to collect users' opinions on customized ASMR generation services.

Expectations on AI-based ASMR Audio Customization Users can easily be satisfied with the visual content of the ASMR video because their choice of whether to watch a video or not is greatly influenced by a thumbnail of this video. However, the mere fact that users can be satisfied with visual content does not imply that they will be satisfied with audio content as well since ASMR video style can vary greatly depending on the content creator. With a generative AI solution, this can be compensated by generating

audio content that suits the user's preferences as well as the visual semantics of a given ASMR video. To support this feature, however, an investigation of user satisfaction with the quality of AI-created ASMR sound should be preceded. In this regard, we investigated whether the AI-generated audio content could replace the real audio content after having revealed that the content used in RQ1 consisted of AI-generated and real audio content. As a result, 57% of the participants responded positively. Participants with positive feedback mentioned that *"I thought audio content from both sides matched with the video quite well!"*, *"There is a possibility of replacement since I could not guess it was AI-generated, but not sure to what extent it can satisfy people?"* Conversely, some participants who gave negative responses pointed out that *"Sound did not naturally match with video in some parts and audio sometimes sounded too sharp"*, *"I don't want to hear audio not matching my expectations!"* Moreover, five of the participants who responded negatively further mentioned that improving sound quality and synchronization with video content should be required to improve user satisfaction.

The participants were also asked to provide their opinions regarding the willingness to use the AI-based ASMR sound customization service if provided. The survey result showed that 78.6% of participants expressed positive responses. Some commented that *"It is usually hard to find such content that fits myself one by one and it would be convenient to have such a function :)"* and pointed out that *"If content that could help people study or fall asleep is made, I would definitely use it!"*; also there was an opinion voiced that *"It is really important to provide audio content similar to real audio"*. Interestingly, all the participants who chose "No" were people who do not usually watch ASMR. Some of them gave answers like *"Well, I don't know... because I don't usually listen to ASMR :("*.

Text-to-ASMR The final objective of our study is to realize a fully automatic AI-based service that can generate ASMR content including both video and sound from the user's text input. Compared to the audio customization service described above which takes the original ASMR video as input, in this case, the service just needs to take the text (sentence) as input to create ASMR content. As a preliminary evaluation, we generated a demo version of an AI-created ASMR video from user text and then asked the participants to provide feedback. For this, we used a text2video service called CogVideo to create a video from the text 'the firewood is burning', and then generated the corresponding sound using our framework to produce the final ASMR video content (Fig. 6).

The participants were asked if they have the intention of using a service generating ASMR content with a reference to an input keyword/text after watching the aforementioned demo video. As a result, 54.8% of the participants said they do. The participants who responded positively mentioned that *"If the output content is highly related to the intention with which the keyword was inputted, it will be really useful"*, *"I would try it because it looks quite a bit interesting! but not sure if I would continue to use it"*. On the contrary, the majority of the remaining 45.2% of the participants who said they were not willing to use it showed negative reactions in terms of the quality of the video, such as *"There seem to be certain limitations to simply creating a video only with text..."* and *"Since the video seems unnatural, I think, users will only listen to the sound :("*. The results reveal that more than half of the participants are interested in and show positive responses for the direction where AI is involved in

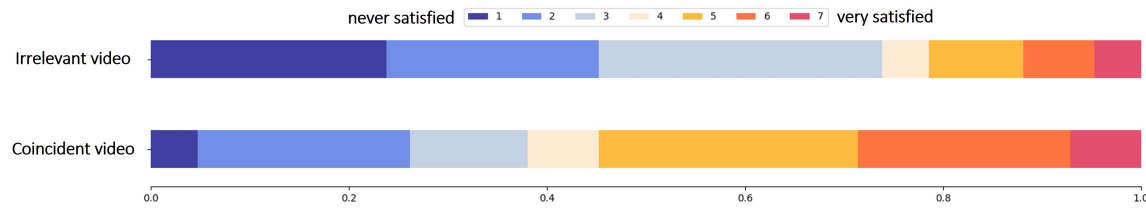


Figure 5: Response to content that consists of sound from irrelevant and coincident video



Figure 6: Demo content generated by Text-to-ASMR technology

content generation, convincing the direction of our future research. A demo video used in our experiment can be found from https://youtu.be/B_ZFdGZXy2Y.

4 CONCLUSION

In this paper, we proposed an AI-based solution that generates ASMR content. Through a user study with 42 participants, we compared ASMR content with AI-generated audio and real audio to figure out if the AI-generated ASMR content can satisfy users, whether AI-generated content sounds natural enough and can be used as normal ASMR content; furthermore, we also learned how visual information affects the user’s satisfaction on ASMR content (RQ1), and investigated user’s opinions on the AI-based ASMR content customization (RQ2).

Through the experiments, we could learn that ASMR sound generated by AI should be well synchronized with the video stream and well reflect visual information since irrelevant sounds significantly degrade user satisfaction. In addition, we found out that noise in the audio also affected the user experience on ASMR content. The overall response to the AI-based ASMR customization was positive. Respondents except those not usually listening to ASMR content mentioned that they had the intention to use the service which generates ASMR audio content matching the user’s tastes from the original ASMR content. Conversely, for text-to-ASMR, many participants questioned the quality of the video. Therefore, it is necessary to generate a more realistic high-quality video as well. As the next step, we will study user-customized ASMR Generation AI services based on our findings.

ACKNOWLEDGMENTS

This work was supported by the NRF grant funded by the Korea Government (MSIT)(2023R1A2C1006300)

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